

## Article

# Business Performance Evaluation for Tourism Factory: Using DEA Approach and Delphi Method

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**Abstract:** The tourism industry contributes more than 10% of global GDP, and creates than 330 million jobs. Since the outbreak of COVID-19, tourism has been one of the hardest hit areas, and one of the most explosive growth sectors, in the post-COVID-19 era. This study analyses the operational efficiency of tourism factories, before and after the COVID-19 outbreak. This study develops a PADME (Product, Aesthetic, Digitalization, Management and Experience) efficiency evaluation model for the non-financial components of tourism factories. This study has also successfully developed the evaluation scale of the PADME model. In addition, with reference to studies on the operational efficiency of financial components, two output variables (turnover and net profit after tax), and three input variables (assets, R&D expenses, and employees) were set, and the efficiency of the PADME model was calculated. The data envelopment analysis (DEA) approach was used to measure the operational efficiency of tourism factories. The empirical research goals of this study are focused on 12 listed companies in Taiwan, with operational efficiency before and after COVID-19 analyzed in relation to their general and individual analyses. The conclusions of this study lead to both enlightening and practical management implications. Academically, this study fills a gap in the research on operational efficiency of tourism factories in the tourism industry.

**Keywords:** tourism factory; creative industry; performance evaluation; data envelopment analysis (DEA); modified Delphi method; case study



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## 1. Introduction

The creative economy has a strong influence in improving competitiveness and drives the development of tourism [1]. The total working population of the service industry in 2020 was 6.879 million, accounting for 59.80% of the total working population in Taiwan, generating 61.52% of Taiwan's Gross Domestic Product (GDP) that year [2]. In 2003, Taiwan's cultural and creative industries were described as community-based cultural and creative industries, while the government's industry-led development termed them general-purpose cultural and creative industries [3].

The creative economy has evolved into one of the drivers of sustainable economic growth based on innovation and creativity [4]. According to Howkins [5], the creative industries employed two million people in the UK, while the financial sector employed only about half as many people as the creative industries. These creative industries are made up of single companies from many related industries. How the corporate performance of creative industry firms is evaluated may be an important reference for the pursuit of sustainability and growth. Most of the research on creative industries has been focused on culture-related studies [6], with few studies examining corporate management.

With 44% of the world's population living in agricultural, rural areas [7], tourism and tourism factories are also an important way of boosting rural economies. The definition given by the Tourism Society is as follows [8]: "Tourism is the temporary, short-term movement of people to destinations outside the places where they normally live and work, and their activities during the stay at each destination. It includes movements for all purposes" [9].

Tourism is the second largest industry in the world; it accounts for more than 10% of global GDP (gross domestic product) [10] and provides employment for 330 million people worldwide [11]. Tourism is considered to be one of the drivers of growth in local economies and the global economy [12–14]. Tourism requires innovation, and tourism is also one of the most competitive sectors for innovation [9]. Innovation is seen as a source of maintaining or gaining competitive advantage and performance in the ever-changing tourism sector [9,15], and tourism factories are one of the innovative products of the tourism industry. In summary, the importance of tourism and creative economy industries is growing.

The purpose of this research is thus to construct a comprehensive performance measurement model for the creative industries at the corporate management level, and to use qualitative and quantitative applied research tools to enable creative industry enterprises to learn from the best in the process of pursuing growth, to enhance the competitiveness of the creative industries. Research [13] suggests that tourism can be used to recover from economic downturns when countries are in financial difficulty, as it has the ability to grow the economy more quickly than other sectors [11]. Today's global economic outlook is more uncertain than ever before [16]. Particularly in times of economic crisis, people will buy goods and services to maintain their basic needs, thereby reducing their travel budgets [16–18]. In general, maintaining high operational efficiency is crucial to the tourism industry; its priority and upmost issue is to measure the operational efficiency of the tourism industry.

After the introduction of Section 1, the study is organized as follows: Section 2 is a literature review. The research method is described in Section 3. The case study, results and discussion are given in Section 4. Section 5 presents the conclusions and future research.

## 2. Literature Review

### 2.1. Creative Industry and Tourism Industry

The creative industry is a development industry involving business, knowledge, and technology in the field of the arts, where human resource creativity is the main area of development [19]. A common definition of the “creative economy” is that it relies on entrepreneurship as a tool for innovation and economic development, and that cities themselves are entrepreneurial in their collective efforts to compete for resources, attract jobs, and attract people [20]. Based on ideas or innovative practices, creative resources such as cultural heritage, skilled workers, and scientific methods, are used to add value to creation [21]. Therefore, the management of creative industries is defined as the use of “sustainable business enterprise management, creativity, and intellectual property” as the main drivers of profits [22]. Creative industries include the creation, production, and distribution of products and services based on “culture” and “creativity”.

The services generated by creative industries and products are heavily dependent on technological development, as they are the drivers of innovation for new technological products [23]. In Indonesia, for example, the contribution of the creative economy to the national economy accounted for 7.44% of GDP in 2016, and the creative economy has become the “local wisdom” of the local culture [21].

The process of digitization of the creative industries, and the rapid development of digital technologies, have accelerated the collaborative innovation networks among creative types of companies. These creative companies are able to collaborate across regions to improve innovation efficiency, reduce production costs, and quickly meet consumers' digital creative needs [24,25]. The digitization of cultural resources combines traditional cultural resources with advanced digital technologies such as virtual reality (VR), augmented reality (AR), artificial intelligence (AI), and 5G to make static cultural resources, that were originally displayed in museums, more active [26].

The concept of One Town One Product (OTOP) originates from the One Village One Product (OVOP) movement in Japan. Dr. Morihiro Hiramatsu proposed that the economic development of each town should be combined with local characteristics, to develop a

handicraft or food specialty industry with local characteristics and differences. The so-called “local” area of local specialty industries is found primarily in townships, towns, and cities, and the specialty products to be developed should have a unique history.

A study [27] by academics found that the volume of research into innovation in tourism was comparable to that assigned into innovation in manufacturing. Related tourism studies have shown the modeling of visitors’ impact on the community [28]. The demand for travel is one of the key factors affecting the performance of the hotel industry [29,30]. A conceptual model is proposed to explain the relationship between key macroeconomic factors and Airbnb providers [31]. In the study [32] related to Airbnb’s share of the economy, a 1% increase in Airbnb revenue reduces the average revenue of a local hotel by 0.016% to 0.031%. Healthy competition in the tourism industry will attract tourists from abroad and make destination tourism more popular with locals [33]. The attractiveness of destination tourism changes over time [34], so the need for constant innovation in tourism makes it important to regularly review the performance of tourism operations.

The topic of agro-tourism has also received a great deal of attention from academics. For example, in wine-producing regions where villages need to take into account their culture and traditions, and where wine tourism in combination with agricultural products is of great importance for the sustainable development of agri-wine [35]. Moreover, when people visit wine regions through tourism opportunities, companies have an opportunity to offer a wide range of services to consumers [36]. An example of an agro-industry, such as grape growing, would offer six major opportunities for tourism development: (1) wine-growing, (2) maintaining a rural lifestyle, (3) landscape conservation, (4) collaboration with government, tourism enterprises and local residents, (5) tourism development, (6) tourism contribution [35].

Responsible tourism is the collective expression of the actions and awareness of all stakeholders towards sustainable travel consciousness [37]. The Triple-A Model (Awareness, Agenda and Action) [38] proposes how to continuously implement the concept of sustainable development, from the three aspects of economy, society, and the environment, with regards to responsible tourism providing advice for development.

Many factories have transformed their manufacturing facilities into tourism factories, or sightseeing factories (known as industrial tourism abroad). This is an effective marketing strategy that provides customers with a sense of intimacy and a direct experience with the brand [39], enhancing corporate brand equity [40]. To become a government-certified tourism factory, the Bureau of Industry (BIE) appraisal process involves organizing experts to evaluate: (1) corporate themes; (2) factory planning and service facilities; (3) facility displays; (4) service quality; (5) the five major components of business management, together with reviewing written information. As of 2021, a total of 144 enterprises with tourism factories in Taiwan have been evaluated. Since tourism factories in the creative industry are a highlight industry that integrates the development of corporate software and hardware resources over a lifetime, this research validates the new model by using tourism factories in the creative industry in Taiwan as its target.

It can be seen that creative economy industries (including tourism factories within the tourism industry) are closely related to socio-economic growth. Previous studies have examined the importance of tourism, the cultural and economic aspects of tourism, the Airbnb business model of tourism and its impact on hoteliers, and destination tourism; but fewer studies on the operational efficiency of tourist factories have been found. In order to meet this void, this study developed a model for measuring the operational efficiency of tourism factories, with financial and non-financial components, for future researchers, operators, and business executives, using Taiwanese companies as the empirical subjects.

## 2.2. Delphi Method

The Delphi method is a research model tool that is based on obtaining “the most reliable consensus of expert opinion” [41,42]. The Delphi method, developed by Helmer and Dalkey in 1950, was an attempt to understand the views of the designers of the former

Soviet strategic plans, to measure the number of atomic bombs needed to paralyze the U.S. munitions industry, and to develop techniques designed to respond to Soviet strategic plans. A series of extensive and intensive questionnaires, supplemented by controlled feedback, were used to synthesize the consensus views of expert specialists to complete this research project, a method known as the Delphi method.

The Delphi method is a systematic and structured group communication process that allows members to fully express their ideas, and communicate on issues or problems, in order to reach consensus [43]. The Delphi method provides a way to gather individual and collective perspectives. Theoharis [44] recommends the Delphi method when researchers want to collect expert judgment for a group decision-making occasion.

Ko and Lu [45] used the modified Delphi method to assess the professional competencies of kitchen staff in cooking, in order to avoid food waste in restaurant kitchens. The modified Delphi method was used to quickly and accurately define the relevant metrics in the study, to evaluate the application of artificial intelligence to sustainable supply chain management performance in the construction material industry.

This research acknowledges the definition and timing of the Delphi method by researchers such as Linstone and Turoff [43] and Theoharis [44] above, and adopts this research method in the development of performance measures for the creative industries.

### 2.3. Performance Assessment

At the most basic level, “performance” is used as a comparison of inputs and outputs. Performance measurement provides a clear picture of the difference between the past and the present, which can be used as a basis for subsequent management [46]. Performance is the purpose of business operations, the very reason for the existence of a business, and is reflected in financial statements as a series of profitability indicators, and in daily operations of the business as output results [47]. Performance is a part of the management control system, and a company can manage its resources more effectively and control its goals when it has a performance measurement and performance management approach [48]. One of the most important steps in performance evaluation is the “establishment of indicators” [49]. Kaplan and Norton [50] were the first to propose the “balanced scorecard” approach to performance evaluation, which includes the following four components: (1) finance, (2) client, (3) internal processes, (4) learning and growth. Since then, performance evaluation has become a multi-faceted assessment of financial and non-financial aspects, which can help a company or organization to develop and operate in the long term, or short term. Performance evaluation is a process of systematically collecting data and comparing it with previously defined criteria in order to assist in the decision-making process, and to assess its efficiency and effectiveness [51]. Evaluating the financial performance of a company is an assessment of the profitability, sustainability, and operational growth of a listed company, which is important for reducing investment risk, safety, and security [52]. Based on the analysis of literature related to corporate performance evaluation, it is known that companies can change and adjust their development strategies corresponding to their corporate status by referring to the results of performance evaluation [53]. The performance of a company’s operations is often measured by purely financial measures in the industry, but tourism factories are influenced by not only financial factors, but also non-financial factors.

### 2.4. DEA

Farrell [54] pioneered the use of multiple inputs and multiple outputs to assess relative efficiency, by replacing a default production function with a non-determined one through a mathematical planning approach. Farrell’s efficiency evaluation model, mentioned in *The Measurement of Productive Efficiency*, is one of the most far-reaching studies in DEA modelling.

### 1. CCR Model

In 1978, Charnes, Cooper, and Rhodes [55] modified the Farrell efficiency model into the CCR model, in which the production frontier of the unit under test is calculated through linear planning, and the relative efficiency of each decision-making unit (DMU) can then be calculated. The CCR model is an input-oriented approach to efficiency, which focuses on how many inputs should be used to achieve relative efficiency at current output levels. The CCR model has an output-oriented model, which aims to compare output achievement at the same level of input, and is known as output-based efficiency. In particular, the CCR model calculates the relative efficiency of each DMU by assuming that the production process is of constant return to scale (CRS).

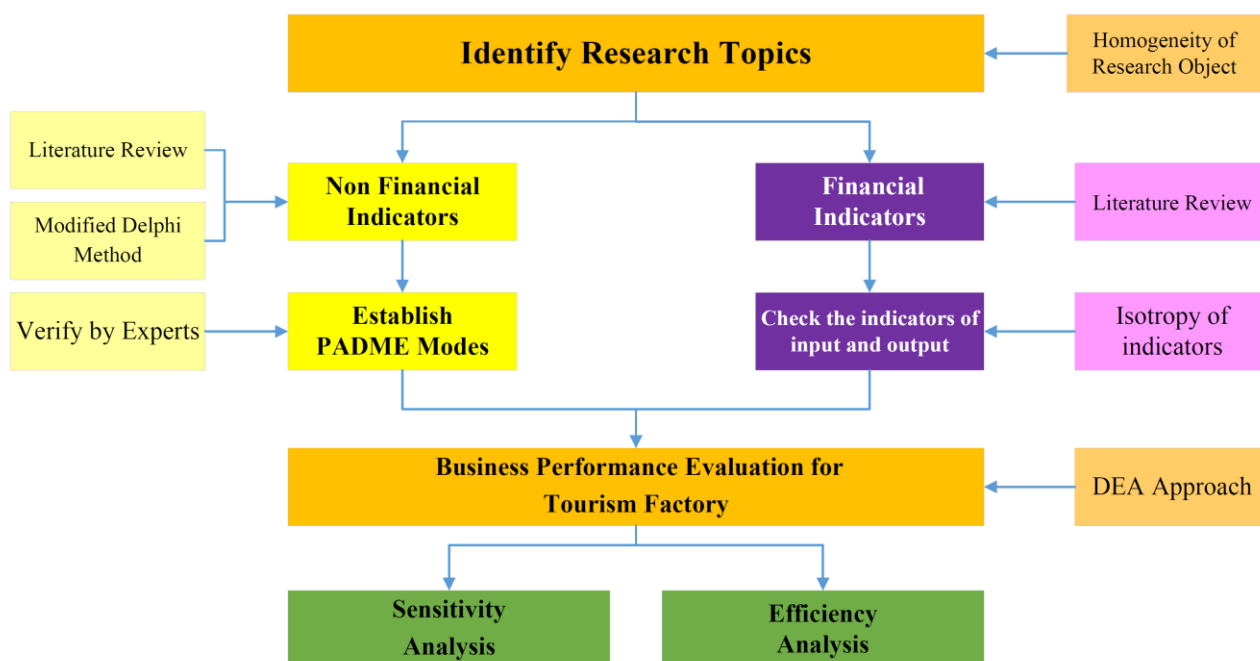
### 2. BCC Model

DMUs may be subject to increasing return to scale (IRS), or decreasing return to scale (DRS), in the production and operation process. Recognizing the return to scale status of DMUs can provide managers with more information on how to improve efficiency [56]. Banker, Charnes, and Cooper [57] suggest that the BCC model can calculate technical efficiency (TE) and scale efficiency (SE), and that the BCC model and CCR model both have input-oriented and output-oriented patterns. The input-based efficiency is calculated when comparing the level of input resources at the same level of output, while the output-based efficiency is calculated when comparing the level of output achieved at the same level of input.

DEA has an important role to play because it does not require the assumption of production functions, or the imposition of subjective weights on multiple inputs and outputs [58,59]. DEA is a tool that can be widely applied in the industry to measure the efficiency of DMUs. It can be used as an objective performance measure for a firm, or a group of firms, as DMUs through the rigorous homogeneity of DMUs and homogeneity of output-input variables, followed by sensitivity analysis [60]. DEA has been used to measure port performance [61], supply chain performance [62], manufacturing efficiency assessment [63], and service sector efficiency assessment [64]. Although this is a very mature application, some industries have yet to make a breakthrough. Emrouznejad and Yan [65], exploring DEA-related articles from 1978 to 2016, and showed that DEA has not received much attention in the creative industries or tourism factories, which do not appear in the top 50 keywords, and is not an industry sector which researchers have investigated to any great depth, highlighting the importance of this study.

### 3. Research Method

This research focuses on tourism factories in the creative industry that have the same characteristics. To construct a measurement model for tourism factories, we used literature discussion and the modified Delphi method to establish Non-Financial Indicators (NFIs) and developed PADME's efficiency model for measuring NFIs. In addition, the literature was used to compile Financial Indicators (FIs) to define input and output measures. The flow chart of this research is shown in Figure 1.



**Figure 1.** Flow chart of the research.

### 3.1. The Development Process of Research Indicators

#### 3.1.1. Literature on Non-Financial Indicators

Digital technology is an important force in promoting the development of virtualization of creative industries. It is necessary to actively promote digital integration in all fields of creative industries; create the core of digital, networked, and intelligent development of creative industries; realize the upgrade and innovation of the whole industrial chain; and promote the improvement of product quality and service experience, to drive higher operating efficiency [66]. Moreover, with the authorization of digital cultural resources, creative enterprises can create and develop a series of digital creative derivative products, activating the online research and development (R&D), production, sales, and consumption ecology of creative industries [67].

According to the United Nations Development Program (UNDP), creative economic development includes arts and crafts, books, films, paintings, festivals, songs, design, digital animation, and audio-visual games [19]. The use of digital transformation will enhance tourism [68]. Tourism is one of the knowledge-intensive industries and the human factor is quite crucial [11].

The creative economy enables tourists and local residents to coexist peacefully, or can even generate memories and visit cultural and historical connotations, thus deepening the connectivity between tourists and local residents with tourism, shaping a deeper understanding of local characteristics [69]. In summary, the above-mentioned research insights identify “product”, “aesthetics”, “digital technology”, and “experience” as important knowledge areas for creative industries. In this research, while developing the performance measurement indicators for creative industries, we first read the literature on creative industries, and after preliminary collection and research, we find that, in addition to the above-mentioned components, the “local characteristics” of local industries should also be considered an important knowledge area of the creative economy.

Hence, the first round of indicators for the non-financial indicators of this research are as follows: product force, aesthetic force, digital force, management force, experience force, and local force.

### 3.1.2. Development of PADME Model by the modified Delphi Method

#### 3. Application of the modified Delphi method

In this study, a modified Delphi method was used to set the non-financial measures using a workout with experts, which traditionally requires more than three rounds of communication with experts to obtain expert consensus. In this study, the modified Delphi method was used to ensure that a high level of consensus was reached in 1–2 rounds with experts. In addition, during the communication process with the experts, the literature on the non-financial aspects of the study was compiled for the experts to browse, and the modified Delphi method was successfully applied to obtain measures of the non-financial aspects of the tourism factory and its scale questions.

Experts must meet the following five conditions: (1) be interested in the application of the modified Delphi method, (2) have the expertise to share knowledge, (3) have expertise in the field or its techniques, (4) have expertise in the subject matter, (5) be able to reach a consensus on the conclusions of the survey [62,70]. Experts must have greater knowledge and experience than the general population and be able to accurately present knowledge standards, reliability, and accuracy, and their judgments must be closer to reality than those of the general population [71]. Researchers have indicated that expert opinion with more than five participants can be used as a reference for analysis [62]. In this research we consulted seven experts, including four with Ph.D. degrees, three with master's degrees, and with an average of 24.6 years of work experience, each of whom was a university teacher or senior manager of a company, with considerable expertise in the creative industry. The expert profiles are shown in Table 1.

**Table 1.** Summary of the basic information of experts in the revised Delphi method.

| Item    | Gender    |           | Education |           | Working Years | Job                 |                             |
|---------|-----------|-----------|-----------|-----------|---------------|---------------------|-----------------------------|
|         | Male      | Female    | Master    | Ph.D.     | Average       | University Teachers | Senior Commercial Executive |
| Content | 4 persons | 3 persons | 3 persons | 4 persons | 24.6 Years    | 2 persons           | 5 persons                   |

The research initially set six dimensions (product power, aesthetic power, digital power, business power, experience power, and local power) and required five experts to complete the questionnaire. In the first round of data exchange, after collecting the experts' questionnaires, we found that the experts disagreed with the local dimension, and that their disagreement votes were greater than their agreement votes. Therefore, in the second round, the local dimension was removed, leaving only (1) product, (2) aesthetics, (3) digital, (4) management, (5) experience. The experts "agreed" on all five dimensions, and each dimension had four questions to support the score of the dimension. Finally, the technical development of this research applied the Delphi method to determine that the measurement of creative industry performance should be composed of five major components: (1) Product Force, (2) Aesthetic Force, (3) Digitalization Force, (4) Management Force, (5) Experience Force. Each of these dimensions is composed of four measurements. The model used in this research is called the "PADME model" (Figure 2).

#### 4. PADME Model

The PADME model, which was developed to measure the performance of the non-financial components of tourism factories, is scored as follows: each component has four questions, and each question is scored using a Likert-type five-point scale, with 1 being the lowest score and 5 being the highest score.

- Product Force: Four questions, each scored from 1 to 5 points. The maximum score for this component is 20 points. The Product Force profile score is designated  $S_P$ .
- Aesthetic Force: Four questions, each scored from 1 to 5 points. The maximum score for this component is 20 points. The Aesthetic Power profile score is designated  $S_A$ .

- Digitalization Force: Four questions, each scored from 1 to 5 points. The maximum score for this component is 20 points. The Digitalization Force profile score is designated  $S_D$ .
- Management Force: Four questions, each scored from 1 to 5 points. The maximum score for this component is 20 points. The Management Force profile score is designated  $S_M$ .
- Experience Force: Four questions, each scored from 1 to 5 points. The maximum score for this component is 20 points. The Experience Force profile score is designated  $S_E$ .

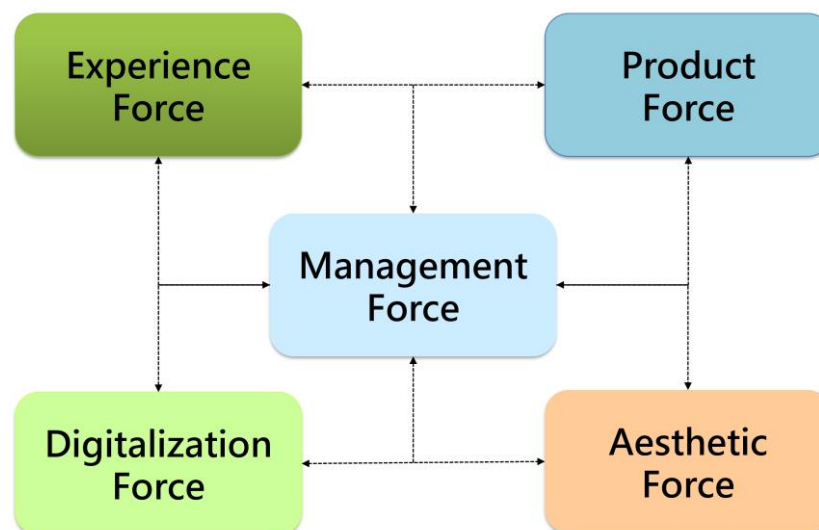


Figure 2. PADME Model.

Total PADME score is 100 points for the non-financial component of the observational factory performance ( $SCORE_{PADME}$ ; designated  $S_{PADME}$ ). The Questionnaire for assessing the PADME model is shown Table 2.

$$S_{PADME} = S_P + S_A + S_D + S_M + S_E \quad (1)$$

Table 2. The questionnaire for assessing the PADME model.

| Item | Measurement Structure | Measuring Items                                                                                             | Rating (5 Points Max.)                                                                                                                 | Sub-Total of Structure Score |
|------|-----------------------|-------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| 1    | Product Force         | Quality of production process                                                                               | <input type="checkbox"/> 1 <input type="checkbox"/> 2 <input type="checkbox"/> 3 <input type="checkbox"/> 4 <input type="checkbox"/> 5 | <input type="checkbox"/>     |
|      |                       | Features of the product (technology/creativity/cultural application)                                        | <input type="checkbox"/> 1 <input type="checkbox"/> 2 <input type="checkbox"/> 3 <input type="checkbox"/> 4 <input type="checkbox"/> 5 |                              |
|      |                       | Showing the content of the story (place + culture)                                                          | <input type="checkbox"/> 1 <input type="checkbox"/> 2 <input type="checkbox"/> 3 <input type="checkbox"/> 4 <input type="checkbox"/> 5 |                              |
|      |                       | Product with servicing package (logistics monitoring capabilities ..., easy to be visible, to find, to buy) | <input type="checkbox"/> 1 <input type="checkbox"/> 2 <input type="checkbox"/> 3 <input type="checkbox"/> 4 <input type="checkbox"/> 5 |                              |
| 2    | Aesthetic Force       | Environmental element 2S (Organizing, cleaning)                                                             | <input type="checkbox"/> 1 <input type="checkbox"/> 2 <input type="checkbox"/> 3 <input type="checkbox"/> 4 <input type="checkbox"/> 5 | <input type="checkbox"/>     |
|      |                       | Aesthetics and Corporate History as well as Local Connections (Local Style)                                 | <input type="checkbox"/> 1 <input type="checkbox"/> 2 <input type="checkbox"/> 3 <input type="checkbox"/> 4 <input type="checkbox"/> 5 |                              |
|      |                       | Corporate Performance and Local Connections                                                                 | <input type="checkbox"/> 1 <input type="checkbox"/> 2 <input type="checkbox"/> 3 <input type="checkbox"/> 4 <input type="checkbox"/> 5 |                              |
|      |                       | The extent of aesthetic innovation and its application                                                      | <input type="checkbox"/> 1 <input type="checkbox"/> 2 <input type="checkbox"/> 3 <input type="checkbox"/> 4 <input type="checkbox"/> 5 |                              |

Table 2. Cont.

| Item | Measurement Structure | Measuring Items                                                                     | Rating (5 Points Max.)                                                                                                                 | Sub-Total of Structure Score |
|------|-----------------------|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| 3    | Digitalization Force  | Implementation of digital media (Line, WeChat, Weibo, IG, FB...)                    | <input type="checkbox"/> 1 <input type="checkbox"/> 2 <input type="checkbox"/> 3 <input type="checkbox"/> 4 <input type="checkbox"/> 5 | <input type="checkbox"/>     |
|      |                       | Extent of upward (supply-side) or downward integration (client-side) development    | <input type="checkbox"/> 1 <input type="checkbox"/> 2 <input type="checkbox"/> 3 <input type="checkbox"/> 4 <input type="checkbox"/> 5 |                              |
|      |                       | Digital development of accumulated community management capabilities                | <input type="checkbox"/> 1 <input type="checkbox"/> 2 <input type="checkbox"/> 3 <input type="checkbox"/> 4 <input type="checkbox"/> 5 |                              |
|      |                       | Diversified integration of digital innovations (e.g., 5G applications...)           | <input type="checkbox"/> 1 <input type="checkbox"/> 2 <input type="checkbox"/> 3 <input type="checkbox"/> 4 <input type="checkbox"/> 5 |                              |
| 4    | Management Force      | The value of overall image performance                                              | <input type="checkbox"/> 1 <input type="checkbox"/> 2 <input type="checkbox"/> 3 <input type="checkbox"/> 4 <input type="checkbox"/> 5 | <input type="checkbox"/>     |
|      |                       | Team Service Capabilities and Results                                               | <input type="checkbox"/> 1 <input type="checkbox"/> 2 <input type="checkbox"/> 3 <input type="checkbox"/> 4 <input type="checkbox"/> 5 |                              |
|      |                       | Business growth capacity (past and future)                                          | <input type="checkbox"/> 1 <input type="checkbox"/> 2 <input type="checkbox"/> 3 <input type="checkbox"/> 4 <input type="checkbox"/> 5 |                              |
|      |                       | Profitability of business operations                                                | <input type="checkbox"/> 1 <input type="checkbox"/> 2 <input type="checkbox"/> 3 <input type="checkbox"/> 4 <input type="checkbox"/> 5 |                              |
| 5    | Experience Force      | Adequate supporting area (beauty and comfort...)                                    | <input type="checkbox"/> 1 <input type="checkbox"/> 2 <input type="checkbox"/> 3 <input type="checkbox"/> 4 <input type="checkbox"/> 5 | <input type="checkbox"/>     |
|      |                       | Aesthetic creativity and innovative application of the field                        | <input type="checkbox"/> 1 <input type="checkbox"/> 2 <input type="checkbox"/> 3 <input type="checkbox"/> 4 <input type="checkbox"/> 5 |                              |
|      |                       | Experience of interaction level and associate customer feedback (one-way / two-way) | <input type="checkbox"/> 1 <input type="checkbox"/> 2 <input type="checkbox"/> 3 <input type="checkbox"/> 4 <input type="checkbox"/> 5 |                              |
|      |                       | Experience evaluation system and adjustment                                         | <input type="checkbox"/> 1 <input type="checkbox"/> 2 <input type="checkbox"/> 3 <input type="checkbox"/> 4 <input type="checkbox"/> 5 |                              |
|      |                       | Total Score                                                                         | <input type="checkbox"/>                                                                                                               |                              |

### 3.2. Research Goals

This research originally planned to use 13 companies listed on the Taiwan Stock Exchange with tourism factories as its research objects, with the relevant operating data of the stock market as observations for analysis. Table 3 below is a list of the 13 tourism factories and their affiliated companies. These 13 companies are all tourism factories, and their products are all tangible commodities, excluding companies in the service industry. Based on the estimated range of the capital of these 13 companies, the capital of Formosa Plastics Co. (NT\$63.65 billion) is much higher than the upper limit of the 99% confidence interval (about NT\$21.5 billion) of the capital of these 13 companies. Since Formosa Plastics Company and the other 12 companies are not homogeneous, Formosa Plastics Company was excluded from this research.

Table 3. List of 12 tourism factories and their affiliated companies.

| DMU              | Name of Tourism Factor                      | Name of Company                       |
|------------------|---------------------------------------------|---------------------------------------|
| DMU <sub>1</sub> | Pineapple mini-cake Factory                 | Vigor Kobo Co., Ltd.                  |
| DMU <sub>2</sub> | Green Future Unlimited GRX™                 | Green Future Unlimited GRX™           |
| DMU <sub>3</sub> | Daxi Tea Factory                            | Taiwan Tea Corporation                |
| DMU <sub>4</sub> | Republic of Chocolate                       | Hun Ya Foods Co., Ltd.                |
| DMU <sub>5</sub> | Grape King Health and Vitality Power Center | Grape King Biomedical Co., Ltd.       |
| DMU <sub>6</sub> | Chi-Sheng Beauty Factory                    | Chi-Sheng Pharma & Biotech Co., Ltd.  |
| DMU <sub>7</sub> | Champion Green Vision Living Aesthetics     | Champion Building Materials Co., Ltd. |

Table 3. Cont.

| DMU               | Name of Tourism Factor                | Name of Company                          |
|-------------------|---------------------------------------|------------------------------------------|
| DMU <sub>8</sub>  | Yulon Experience Center               | Yulon Automobile Manufacturing Co., Ltd. |
| DMU <sub>9</sub>  | Taiyen Tongxiao Tourism Factory       | TaiYen Co. Inc.                          |
| DMU <sub>10</sub> | Kangnaxiang Nonwoven Creative Kingdom | K N H Enterprise Co., Ltd.               |
| DMU <sub>11</sub> | Eminent Interactive Luggage World     | Eminent Luggage Co., Ltd.                |
| DMU <sub>12</sub> | Li-Kang Health Tourism Factory        | Li-Kang Biomedical Co., Ltd.             |

Note: Formosa Biomedical Health Center in Formosa Plastics Co., Ltd. is not the object in this research.

### 3.3. Variables for Inputs and Outputs

#### 1. Input variables

Financial performance is a method of assessing the efficiency of financial performance at the management level of a business [72]. Developing human resources helps to improve the tangible and intangible financial performance of a company [73,74]. Total assets are a critical indicator in assessing the performance of a company [60]. Research and development (R&D) expenses need to be included as a measure of a company's performance in order to provide a more realistic picture of the relevance of R&D investment to real output [75]. Based on the above studies, three indicators were used as input variables for this research: total assets, research and development (R&D) expenses, and the number of employees in the business. The definitions of the input variables are given in Table 4.

Table 4. Introduction to the input variables and their definitions.

| Item | Indicators   | Code  | Definitions                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------|--------------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Assets       | $X_A$ | Total assets represent all assets owned or controlled by an enterprise. It includes current assets, long-term investments, fixed assets, intangible and deferred assets, and other long-term assets. It is the total amount of assets on an enterprise's balance sheet.                                                                                                                                                                                              |
| 2    | R&D Expenses | $X_R$ | Expenses on research into new products, technologies, services or creations, improved production techniques, improved labor supply techniques and improved manufacturing processes for the profit-making business.                                                                                                                                                                                                                                                   |
| 3    | Employees    | $X_L$ | Individuals (whether native or foreign) who provide labor for the enterprise under the supervision of the enterprise, excluding those who have only a contractual relationship with the enterprise (e.g., insurance salesmen who only earn commissions, are paid upon completion of their contracted work, and do not enjoy the rights and benefits of employees as provided for by law), outsourced or dispatched employees; and excluding directors (supervisors). |

#### 2. Output variables

The assessment of a company's operating performance requires consideration of its net revenue and profit after tax, which are important output indicators in the evaluation of financial components [76]. In addition, the efficiency value developed in this research to measure non-financial components is one of the output variables, and is therefore referred to as  $S_{PADME}$ . The introduction to the output variables and their definitions in this research are shown Table 5.

**Table 5.** Introduction to the output variables and their definitions.

| Item | Indicators           | Code        | Definitions                                                                                                                                                       |
|------|----------------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Turnover             | $Y_T$       | Net revenue is calculated by deducting sales returns and discounts, i.e., the amount of consumer returns, discounts, etc.                                         |
| 2    | Net Profit after Tax | $Y_P$       | The retention of a company's profit after the required income tax has been paid on the total profit, also commonly referred to as profit after tax or net income. |
| 3    | Score of PADME       | $S_{PADME}$ | Expert assessment scores for the five major components: Product Force, Aesthetic Force, Digitalization Force, Management Force, Experience Force.                 |

### 3. Definition of Production Functions for the Tourism Factory

The traditional production function for measuring the efficiency of a decision-making unit (DMU) purely in terms of financial related indicators, is two output variables ( $Y_T$  and  $Y_P$ ) and three input variables ( $X_A$ ,  $X_R$  and  $X_L$ ), modelled in Equation (2).

$$Y_T + Y_P = f(X_A + X_R + X_L) \quad (2)$$

$$Y_T + Y_P \leq 1$$

In this research, a PADME model was developed to measure the operational efficiency of the non-financial components of creative industry tourism factories. The efficiency value ( $S_{PADME}$ ) is calculated as in Equation (1) and the production function model to measure the overall operational efficiency of creative industry tourism factories is given by Equation (3).

$$Y_T + Y_P + S_{PADME} = f(X_A + X_R + X_L) \quad (3)$$

$$Y_T + Y_P + S_{PADME} \leq 1$$

In this research, there are three inputs and three outputs for a total of six indicators. Since Golany and Roll [77] suggested that when using the DEA method to evaluate the efficiency value of each DMU, the number of DMUs should be at the least two times the total number of indicators, the number of DMUs in this research should be equal to, or greater than, 12.

### 4. Data source

This research uses the Taiwan Stock Market Observation Post System to derive the input and output index values of the 12 companies in research from 2019 and 2020.

### 5. Correlation analysis of output and input variables

There are three output variables and three input variables in this research. The output variables are turnover (code  $Y_T$ ), net profit after tax (code  $Y_P$ ) and score of PADME (code  $S_{PADME}$ ). The input variable are assets (code  $X_A$ ), R&D expenses (code  $X_R$ ) and employees (code  $X_L$ ). In this research, the isotropy between the output and input variables was checked and a Pearson correlation analysis was performed using SPSS 26 software, as shown Table 6.

**Table 6.** Pearson's correlation analysis.

| Output Variables | Input Variables | $X_A$    | $X_R$    | $X_L$    |
|------------------|-----------------|----------|----------|----------|
| $Y_T$            |                 | 0.954 ** | 0.650 ** | 0.702 ** |
| $Y_P$            |                 | 0.647 ** | 0.481 *  | 0.472 *  |
| $S_{PADME}$      |                 | 0.367    | 0.439 *  | 0.636 ** |

Note: \*\*. The correlation is significant at the 0.01 level (two-tailed). \*. The correlation is significant at the 0.05 level (two-tailed).

The correlation analysis results show that the three output variables are all correlated with the three input variables, and only the output variable  $S_{PADME}$  does not show a significant correlation with the input variable  $X_A$ . This phenomenon is the same as the actual situation, representing the non-financial indicator  $S_{PADME}$ . There is no significant correlation between the score of DMU and the amount of assets of DMU; it is not necessary to have high assets to have a higher score for non-financial purchases. Therefore, the output variables and input variables used in this research passed the isotropy test.

#### 4. Results and Discussion

The production function derived from Equation (3) of the tourism factory measurement model constructed in this research has three input variables and three output variables. The DEA method is used with six variables to measure the performance of tourism factory operational efficiency in the creative industry. This section presents the efficiency analysis and the sensitivity analysis, and their management implications.

##### 4.1. Efficiency Analysis

###### 1. General analysis

In this research, 12 listed companies with tourism factories were analyzed by deriving operational data from their annual reports. After the data collection was completed, the technical efficiency (TE), pure technical efficiency (PTE), and scale efficiency (SE) of the 12 DMUs were obtained using DEAP 2.1 software. Table 7 below shows the efficiency values of each DMU for the 12 DMUs in 2019, the first year of the COVID-19 outbreak, and in 2020.

**Table 7.** Efficiency values for the 12 DMUs in the two years following the COVID-19 outbreak.

| DMU               | Year | 2019  |       |       | 2020  |       |       |
|-------------------|------|-------|-------|-------|-------|-------|-------|
|                   |      | TE    | PTE   | SE    | TE    | PTE   | SE    |
| DMU <sub>1</sub>  |      | 1.000 | 1.000 | 1.000 | 0.176 | 1.000 | 0.176 |
| DMU <sub>2</sub>  |      | 0.821 | 0.845 | 0.972 | 0.972 | 1.000 | 0.972 |
| DMU <sub>3</sub>  |      | 0.339 | 0.363 | 0.933 | 1.000 | 1.000 | 1.000 |
| DMU <sub>4</sub>  |      | 0.886 | 1.000 | 0.886 | 1.000 | 1.000 | 1.000 |
| DMU <sub>5</sub>  |      | 0.588 | 1.000 | 0.588 | 1.000 | 1.000 | 1.000 |
| DMU <sub>6</sub>  |      | 0.875 | 0.951 | 0.921 | 1.000 | 1.000 | 1.000 |
| DMU <sub>7</sub>  |      | 0.480 | 1.000 | 0.480 | 0.781 | 0.949 | 0.824 |
| DMU <sub>8</sub>  |      | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 |
| DMU <sub>9</sub>  |      | 0.623 | 1.000 | 0.623 | 0.699 | 0.953 | 0.734 |
| DMU <sub>10</sub> |      | 0.666 | 0.811 | 0.822 | 1.000 | 1.000 | 1.000 |
| DMU <sub>11</sub> |      | 0.909 | 1.000 | 0.909 | 0.424 | 0.971 | 0.437 |
| DMU <sub>12</sub> |      | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 |
| Mean              |      | 0.766 | 0.914 | 0.845 | 0.769 | 0.988 | 0.777 |

Table 7 shows a significant reduction in scale efficiency (SE) in 2020, clearly due to the impact of COVID-19 on the scale of operation of the DMU<sub>s</sub>. For pure technical efficiency (PTE), the second year of the epidemic showed an increase in PTE, as the DMU<sub>s</sub> maintained a mentality of continuous improvement in their operations despite the effect of the epidemic on their business, and increased their technology-related investments and output. This also reflects the fact that these tourism factory companies did not react negatively to the epidemic, but rather that the epidemic prompted them to strengthen their technological development.

## 2. Individual analysis

Table 7 shows that DMU<sub>1</sub> and DMU<sub>11</sub> were more strongly affected by the epidemic, resulting in a significant reduction in the respective efficiency values from the previous year. In addition, for the DMU<sub>s</sub> that were profitable for two consecutive years, DMU<sub>12</sub> had the best performance with a relative efficiency of 1.0 for both years. This shows that biotech companies were not negatively impacted by the outbreak of COVID-19, and their performance is among the best in the tourism factory category.

### 4.2. Sensitivity Analysis

To ascertain whether the sensitivity of the research population had any impact on the performance evaluation, the data of DMU<sub>11</sub> and DMU<sub>12</sub> with the highest number of peer counts in 2019 were removed from the analysis. The efficiency values of the remaining 10 DMU<sub>s</sub> were then assessed using the DEA method as shown Table 8. In the same operation, the data of DMU<sub>6</sub> with the highest number of peer counts in 2020 were first removed, and the efficiency of the remaining 11 DMU<sub>s</sub> was then evaluated using the DEA method. The scale efficiency (SE) values for both years of the sensitivity test showed a decreasing trend, and the SE values for each DMU did not change much from the efficiency values obtained before the sensitivity test (Table 9), which means that these DMU<sub>s</sub> did not have a significant impact on the evaluation of the efficiency values, and therefore passed the sensitivity analysis.

**Table 8.** Number of times each DMU<sub>s</sub> was peer counted in both years.

| DMU               | Year | 2019        | 2020        |
|-------------------|------|-------------|-------------|
|                   |      | Peer Counts | Peer Counts |
| DMU <sub>1</sub>  |      | 2           | 2           |
| DMU <sub>2</sub>  |      | 0           | 0           |
| DMU <sub>3</sub>  |      | 0           | 0           |
| DMU <sub>4</sub>  |      | 0           | 0           |
| DMU <sub>5</sub>  |      | 0           | 0           |
| DMU <sub>6</sub>  |      | 0           | 4           |
| DMU <sub>7</sub>  |      | 0           | 0           |
| DMU <sub>8</sub>  |      | 2           | 3           |
| DMU <sub>9</sub>  |      | 0           | 0           |
| DMU <sub>10</sub> |      | 0           | 1           |
| DMU <sub>11</sub> |      | 3           | 0           |
| DMU <sub>12</sub> |      | 3           | 0           |

**Table 9.** The result of sensitivity analysis using DEAP 2.1 software.

| DMU | Year              | 2019  | 2020  |
|-----|-------------------|-------|-------|
|     |                   | SE    | SE    |
|     | DMU <sub>1</sub>  | 1.000 | 0.176 |
|     | DMU <sub>2</sub>  | 0.985 | 0.998 |
|     | DMU <sub>3</sub>  | 0.433 | 0.183 |
|     | DMU <sub>4</sub>  | 0.932 | 1.000 |
|     | DMU <sub>5</sub>  | 0.664 | 1.000 |
|     | DMU <sub>6</sub>  | 1.000 | -     |
|     | DMU <sub>7</sub>  | 0.480 | 0.815 |
|     | DMU <sub>8</sub>  | 1.000 | 1.000 |
|     | DMU <sub>9</sub>  | 0.623 | 0.731 |
|     | DMU <sub>10</sub> | 0.719 | 1.000 |
|     | DMU <sub>11</sub> | -     | 0.433 |
|     | DMU <sub>12</sub> | -     | 1.000 |
|     | Mean              | 0.784 | 0.758 |

#### 4.3. Management Implications

When a company is affected by uncontrollable factors or a relatively poor environment, such as COVID-19, it would be distorting to assess operational performance only from financial data, or other methods that require subjective weighting. The integrated PADME assessment model developed in this research can provide a more realistic picture of pure technical efficiency, which is the productivity of a company due to management and technical factors. In practical terms, the PADME scores can also provide insight into which of the non-financial components the company scores lowest, and can be used as an important reference for continuous self-improvement.

## 5. Conclusions and Future Research

### 5.1. Conclusions

Taking the financial crisis of 2008 as an example, international tourist arrivals and international tourism revenues fell by 4% and 6%, respectively [16,78]. The extent of the international impact of the epidemic, since its outbreak in 2019, is difficult to estimate [79] and is even greater than the 2008 financial crisis, making it all the more important for us to respond calmly in the current post-COVID-19 era.

This research is a comprehensive, innovative, and integrated application of qualitative and quantitative tools. It first applies the knowledge and power of the modified Delphi method to develop non-financial indicators and measurement models for measuring tourism factories in creative industries. Then, using the financial and non-financial measures of creative tourism factories, this study constructs a model that is consistent with the operational efficiency of creative tourism factories. The results of this research will enable companies to identify which of their financial or non-financial components are less efficient, and to identify benchmarks for sustainable growth.

To address a research gap, the first contribution of this research is to take the lead in establishing an operational efficiency assessment model for creative industry tourism factories. In addition, using the DEA method, in which pure technical efficiency is the production efficiency of enterprises due to management, technology, and other factors, this research confirms the practicality of the new model of this research and development through empirical analysis, and evaluates the non-financial configuration efficiency value of creative industry tourism factories through the newly constructed PADME measurement model. The pure technical efficiency of the DMU may be accurately measured, and the

comprehensive technical efficiency measured by the DEA is more accurate, which is the second contribution of this research.

As a result of the epidemic, fewer tourism factories in Taiwan have been able to maintain normal operations, compared to before the epidemic, especially those that have been able to continue operating in the post- COVID-19 era. Therefore, this study is based on tourism factories listed in Taiwan, with the concept of sustainable operation.

1. The PADME measurement model for non-financial components of tourist factories.

Due to the operational characteristics of tourism factories, the operational efficiency of tourism factories cannot be measured solely by financial measures. The PADME model developed in this study, to measure the operational efficiency of non-financial components of tourism factories, has been validated by experts and case studies; therefore, the PADME is operational and usable.

2. The DEA integrated model for measuring the operational efficiency of tourism factories is validated by case studies.

This study proposes that measurement of the operational efficiency of a tourism factory should be based on the integration of the efficiency of the financial and non-financial components. The DEA method allows for a fair assessment of operational efficiency without the need to set artificial weights, which is valuable in practical terms.

3. Major changes in the operating strategies of tourism factory enterprises before and after the outbreak.

During COVID-19, the operations of non-biotech tourist factory enterprises declined significantly. The efficiency values of the tourism factories before and after the epidemic show that the pure technical efficiency (PTE) increased from 0.914 before the epidemic, to 0.988 after the epidemic, and that the enterprises were able to adjust to the impact of the epidemic, so that they could engage in R&D activities to prepare for the post-epidemic period. At the same time, the scale efficiency (SE) of enterprises also decreased sharply, from 0.845 to 0.777, which shows that enterprises slowed down their expansion during the epidemic in order to preserve their capital and strength, so as to build up their capital, and then take the initiative when the dawn breaks to face the light without the epidemic.

4. Government policies should take into account the differences in industry sectors.

For example, although tourism factories are unable to attract tourists, encouraging the private sector to purchase from tourism factories on integrated e-commerce platforms is a horizontal integration that cannot be done by the industry alone.

## 5.2. Future Research

Tourism is one of the most energy-intensive industries [80]. Approximately 8% of the total global greenhouse gas (GHG) emissions is generated by tourism [81]. Environmental pollution and climate change can have a serious impact on tourism [82]. The demand for energy is increasing rapidly; energy is considered one of the key issues in the world [83,84]. This study did not include the issue of energy as a non-operational performance component of this study, and we suggest that subsequent studies could continue to strengthen the tourism factory in relation to energy and environmental issues.

Since creative industries are smaller in scale than traditional manufacturing industries, it is not easy to collect data from the creative industry. However, in this research, DMU<sub>s</sub> with twice the number of input and output variables were used to conduct the research, and we were not restricted from conducting the research. The target population of this research is limited to publicly traded tourism factories in Taiwan, which may have a better operating quality than small and medium-sized tourism factory enterprises. It is suggested that subsequent studies could be conducted to extend the scope of performance evaluation of creative industries by expanding this research methodology to small and medium-sized tourism factories, as well as a comparison of different regions.

The implementation of CSR by enterprises helps to increase the popularity of the destination and its attractiveness [85–87]. This study has not examined the issue of energy and tourism factories, as some say that tourism is one of the most energy-intensive industries, and some studies suggest that growth in tourism will lead to environmental damage. This study suggests that future research could be directed towards the implementation of CSR in tourism factories to conduct practical and applied research on operational efficiency, so that more tourism companies can see that CSR is not a drain on corporate resources, but will indirectly and concretely drive corporate revenue and profitability, allowing companies to see the benefits of implementing CSR and attracting more companies to the CSR train.

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